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Cactus survey at the Floresta Nacional of Contendas do Sincorá, Bahia, Brazil

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Summary: The Brazilian state of Bahia is home to 98 out of 260 species of the country's native Cactaceae. The Floresta Nacional Contendas do Sincorá (FLONA Sincorá), in the Município de Contendas do Sincorá has vegetation, climate and temperatures typical of the Caatinga biome. The survey intended to increase the knowledge regarding the cacti of protected areas within the state of Bahia and was based on fieldwork carried out in April and July 2014 and February 2015, complemented by studies of herbarium material. The FLONA Sincorá harbours 11 genera and 14 species within its area: *Arrojadoa penicillata* (Gürke) Britton & Rose; *Brasilicereus phaeacanthus* (Gürke) Backeb; *Brasiliopuntia brasiliensis* (Willd.) A. Berger; *Cereus jamacaru* DC.; *Espostoopsis dybowskii* (Rol.-Goss.) Buxb; *Hylocereus setaceus* (Salm-Dyck) R. Bauer; *Melocactus concinnus* Buining & Brederoo; *Pereskia aureiflora* F. Ritter; *Pereskia bahiensis* Gürke; *Pilosocereus catingicola* (Gürke) Byles & G.D. Rowley ssp *catingicola*; *Stephanocereus leucostele* (Gürke) A. Berger; *Tacinga inamoena* (K. Schum.) N.P. Taylor & Stuppy; *Tacinga funalis* Britton & Rose; and *Tacinga palmadora* (Britton & Rose) N.P. Taylor & Stuppy. Of these, *Espostoopsis dybowskii*, *Pereskia bahiensis* and *Stephanocereus leucostele* are endemic to Bahia. The east Brazilian genus *Tacinga* was the largest, with three species. An identification key for all taxa, together with descriptions, illustrations, taxonomic and conservation and threat statements are presented.
Key words: Cactaceae, Caatinga, floristics, protected areas, survey.

Resumo: O estado da Bahia abriga 98 das 260 espécies de Cactaceae nativas ocorrentes no Brasil. A Floresta Nacional Contendas do Sincorá (FLONA Sincorá), localizada no Município de Contendas do Sincorá apresenta vegetação, clima, temperatura e fisionomias características do domínio fitogeográfico da Caatinga. Com o objetivo ampliar o conhecimento da flora local, especificamente das Cactaceae em Áreas Protegidas do estado da Bahia, um levantamento baseado em coletas em abril e julho de 2014 e fevereiro de 2015 e de estudo de material depositado nos herbários foi realizado. Foram registrados 11 gêneros e 14 espécies dentro da área da FLONA Sincorá: *Arrojadoa penicillata* (Gürke) Britton & Rose; *Brasilicereus phaeacanthus* (Gürke) Backeb; *Brasiliopuntia brasiliensis* (Willd.) A. Berger; *Cereus jamacaru* DC.; *Espostoopsis dybowskii* (Rol.-Goss.) Buxb; *Hylocereus setaceus* (Salm-Dyck) R. Bauer; *Melocactus concinnus* Buining & Brederoo; *Pereskia aureiflora* F. Ritter; *Pereskia bahiensis* Gürke; *Pilosocereus catingicola* (Gürke) Byles & G.D. Rowley subsp. *catingicola*; *Stephanocereus leucostele* (Gürke) A. Berger; *Tacinga inamoena* (K. Schum.) N.P. Taylor & Stuppy; *Tacinga funalis* Britton & Rose e *Tacinga palmadora* (Britton & Rose) N.P. Taylor & Stuppy. Das espécies encontradas na região, *Espostoopsis dybowskii*, *Pereskia bahiensis* e *Stephanocereus leucostele* são as únicas endêmicas do estado da Bahia. O gênero *Tacinga*, exclusivo do Leste do Brasil, foi representado por três espécies. Uma chave de identificação para os táxons estudados, juntamente com descrições, ilustrações, comentários

taxonômicos e sobre a situação da conservação e ameaças são apresentados.
Palavras-chave: áreas protegidas, Cactaceae, Caatinga, florística, levantamento.

Zusammenfassung: Der brasilianische Bundesstaat Bahia ist die Heimat von 98 der 260 im Land heimischen Cactaceae. Die Vegetation, das Klima und die Temperaturen des Naturschutzgebietes „Floresta Nacional Contendas do Sincorá" (FLONA Sincorá) in der Gemeinde Contendas do Sincorá sind typisch für das Caatinga-Biom. Eine Bestandsaufnahme hatte das Ziel, das Wissen über die in den Schutzgebieten Bahias vorkommenden Kakteen zu verbessern. Sie basierte auf Felduntersuchungen, die im April und im Juli 2014 sowie im Februar 2015 durchgeführt wurden, ergänzt durch die Untersuchungen von Herbarmaterial. Im Gebiet von FLONA Sincorá kommen 11 Gattungen und 14 Arten vor: *Arrojadoa penicillata* (Gürke) Britton & Rose; *Brasilicereus phaeacanthus* (Gürke) Backeb.; *Brasiliopuntia brasiliensis* (Willd.) A. Berger; *Cereus jamacaru* DC.; *Espostoopsis dybowskii* (Rol.-Goss.) Buxb.; *Hylocereus setaceus* (Salm-Dyck) R. Bauer; *Melocactus concinnus* Buining & Brederoo; *Pereskia aureiflora* F. Ritter; *Pereskia bahiensis* Gürke; *Pilosocereus catingicola* (Gürke) Byles & G. D. Rowley subsp. *catingicola*; *Stephanocereus leucostele* (Gürke) A. Berger; *Tacinga inamoena* (K. Schum.) N. P. Taylor & Stuppy; *Tacinga funalis* Britton & Rose und *Tacinga palmadora* (Britton & Rose) N. P. Taylor & Stuppy. Von diesen sind *Espostoopsis dybowskii*, *Pereskia bahiensis* und *Stephanocereus leucostele* in Bahia endemisch. Die ostbrasilianische Gattung Tacinga ist dabei mit drei Arten die größte Gruppe. Ein Bestimmungsschlüssel für alle Taxa wird zusammen mit Angaben zu Taxonomie, Schutz und Gefährdung präsentiert.
Suchbegriffe: Cactaceae, Caatinga, Floristik, Schutzgebiete, Bestandsaufnahme.

Introduction

With c.124 genera and 1438 species distributed throughout the Americas, Cactaceae is currently divided into four subfamilies primarily on the basis of molecular studies: Cactoideae, Opuntioideae, Pereskioideae and Maihuenioideae (Hunt *et al*., 2006). These subfamilies can be recognized morphologically through the presence or absence of leaves and glochids, general habit and distinctive seed characters.

Eastern Brazil is considered the third most important centre of diversity for the family, including many different vegetation types that harbour cactus species of all subfamilies, excluding Maihuenioideae, which is endemic to southern South America (Taylor, 1997; Taylor & Zappi, 2004).

The present state of knowedge of Cactaceae taxonomy in Brazil differs much from the 13 genera and 121 species treated by Schumann (1890) in the *Flora Brasiliensis*. By 2010, Zappi *et al*. (2010) listed 37 genera and 233 species of which 174 are endemic. Today, 39 genera and 260 native species are recognized, of which 187 are endemic to Brazil. Northeastern Brazil harbours 105 species, with Bahia as the richest state in terms of species numbers, with 98 species (BFG 2015). The Caatinga biome has 63 endemic species, however, deforestation and habitat destruction have directly affected the cactus populations within the Brazilian drylands (Zappi *et al*., 2011).

Despite the fact that there are many widespread Cactaceae species in the Caatinga, habitat degradation impacts on young plant establishment, interferes with the pollination and dispersal vectors, and includes the yet unknown effects from climate change (INSA, 2014), all of which may affect many of the endemic species that are found in narrow ecological niches.

Considering the endemism of eastern Brazilian cacti, it is important to ascertain whether the species are currently found within protected areas (Zappi *et al*., 2011). In terms of threat, 75 cactus species are already recorded in the Brazilian Red List (Martinelli & Moraes, 2013), and habitat and distribution studies are needed to monitor their populations. The aim of this study is to improve the state of knowledge regarding the conservation state of the Cactaceae in the Floresta Nacional Contendas do Sincorá (FLONA Sincorá) by surveying the species and providing identifications and descriptions that enable their easy recognition. Another paper originating from this study, dealing with Cactaceae populations in the same area has been published already (Ribeiro-Silva *et al*., 2016).

Methods

Location

The study area, the Floresta Nacional Contendas do Sincorá – FLONA Sincorá, is located at the Serra do Sincorá, within the municipality of Contendas do Sincorá, in the Brazilian state of Bahia. The Serra do Sincorá occupies the central-eastern edge of the Chapada Diamantina mountain range, between the villages of Afrânio Peixoto (formerly Estiva) to the north and Sincorá Velho to the south, forming a mountain system located between 12° 15' – 13° 45'S and 41° 10' – 41° 30'W (Pedreira, 2002).

The municipality of Contendas do Sincorá is found 20km away from Tanhaçu and occupies a

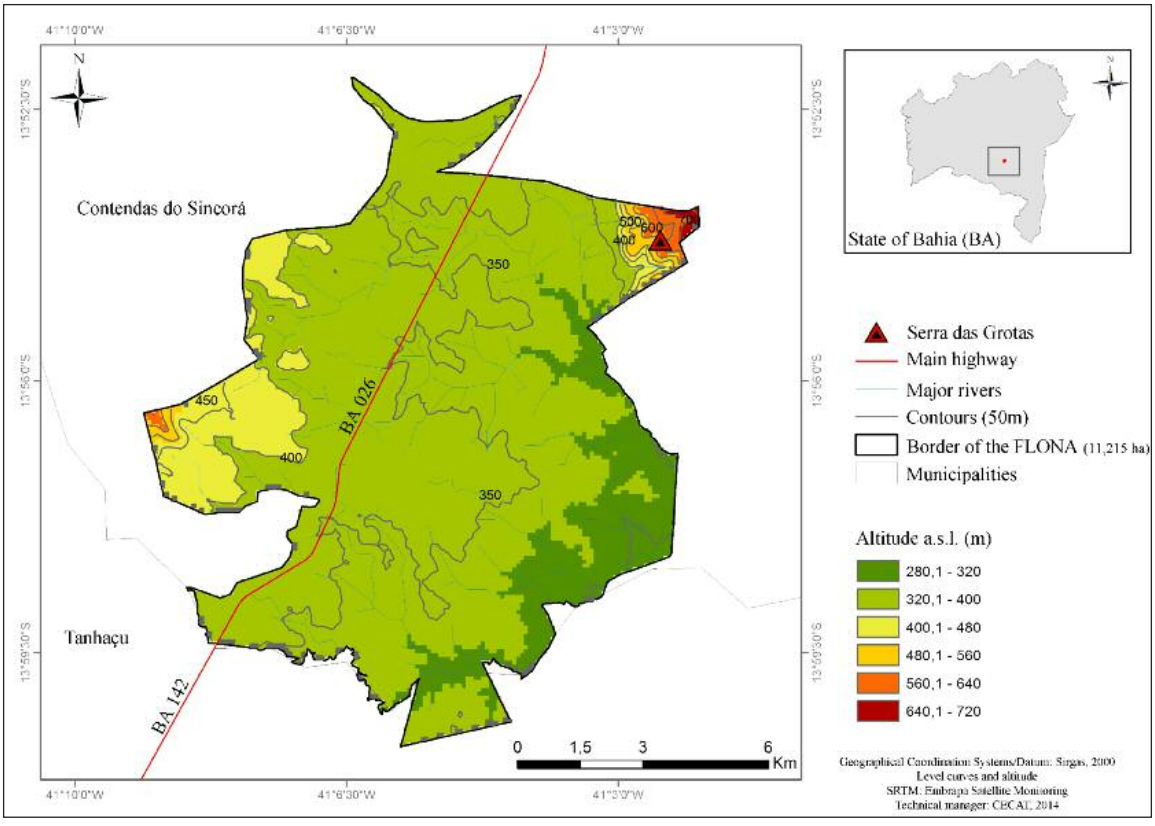


Figure 1. Map of FLONA Sincorá, showing its location within the municipality of Contendas do Sincorá, Bahia, Brazil.

depression between the northern south-Bahian plateau (Planalto de Maracás) and the central-south part of the Chapada Diamantina (Freitas & Borges, 2011).

The Floresta Nacional, or FLONA, falls within a conservation area category known as 'national forest', specifically designed to allow sustainable use of natural resources. This reserve was created in 1999 and measures approx. 11,034 hectares (IBAMA, 2006). The mean temperature in the region is 23°C and relative humidity is between 60–80%. The climate in the area is semi-arid, classified as 'BSwh' according to Köppen (Alvares *et al*., 2013), a hot climate with irregular rainfall (500–700mm) concentrated between November and January. The scarce water availability can be explained by the presence of mountain barriers, namely the Serra das Grotas and the Planalto de Maracás-Jaguaquara to the east and, to the west, the Serra de Cabeça Inchada, blocking the entrance of humid winds that are directed upwards and cause rainfall on the outer slopes, leaving FLONA Sincorá (Figure 1) effectively in a rain shadow situation (IBAMA, 2006).

Located within the Caatinga biome, the FLONA Sincorá includes shrubby and forested scrub and rock outcrops. Near rivers and watercourses taller gallery forest is found (IBAMA, 2006). The shrubby vegetation, referred to in the present study as 'shrubby Caatinga', may be more or less open and is often associated with poorer, more rocky or stony soil. Taller forest with trees sometimes reaching 6m, is here referred to as 'tall Caatinga' and grows on deeper soil and sometimes has an impenetrable under storey with abundant thorny shrubs and lianas.

Floristic survey

Plant collections (herbarium specimens) were made during two trips in 2014 and one in 2015, using the existing tracks in the FLONA Sincorá. Fertile individuals were collected and photographed in the field, documenting also the habitat where the populations were found. Details such as plant height, appearance, and other characters that normally disappear during the herbarium drying process were noted. The material found was identified using specialized literature

and the vouchers were deposited at the Universidade Federal do Recôncavo da Bahia Herbarium (HURB). Material from other herbaria, such as CEN, HUEFS and ALCB (Holmgren *et al*., 1990), was also examined and cited when relevant for this work.

Taxonomic treatment

Cactaceae Juss.

Perennial plants mostly with photosynthetic, succulent stems of varied shape (segmented, flattened, trigonous or cylindric, ribbed or tubercled), sometimes with mucilaginous cells or gummiferous channels; axillary meristems represented by short shoots called areoles which give rise to hairs, glochids, spines, flowers, leaves or new stems. Flowers often solitary, sometimes in panicles, or in complex structures known as cephalia. Flowers generally showy, hermaphrodite, radially symmetric or rarely slightly zygomorphic, receptacular hypanthium or 'pericarpel', often covered with scales and areoles with spines and trichomes, flower tube present or sometimes reduced (*Rhipsalis*); perianth-segments transitioning from external and sepaloid to internal and petaloid; stamens numerous forming one or more continuous spiralled series, inserted within the floral tube, anthers basifixed; nectary disc-shaped above the ovary; ovary inferior, unilocular, placentation basal or parietal, stigma lobed, number of lobes generally equal to the number of fused carpels. Fruits succulent, dehiscent or indehiscent, generally succulent, globose to turbinate, with or without perianth remnants when ripe, pericarp spiny, scaly or smooth, greenish, white or colourful; funicular pulp colourful, solid (*Pilosocereus*), mucilaginous or translucent (tribe Rhipsalideae); seeds commonly numerous, ellipsoid to cochleariform, pear-shaped or globose, testa exposed or covered by a sclerified, bony aril (*Brasiliopuntia*, *Nopalea*, *Opuntia*, *Tacinga*); periclinal cell-walls flat to convex, hilum-micropilar region with two depressions.

A total of 14 species distributed in 11 genera were collected at the FLONA Sincorá. The following species were observed in the region, but outside the boundaries of the protected area: *Arrojadoa rhodantha* (Gürke) Britton & Rose, *Melocactus inconcinnus* Buining & Brederoo, *Pilosocereus gounellei* (F.A.C. Weber) Byles & Rowley, *P. pentaedrophorus* (J.F. Cels) Byles & G.D. Rowley subsp. *robustus* Zappi, and it is possible that *Tacinga werneri* (Eggli) N.P. Taylor & Stuppy may also occur within the reserve. All these species are keyed out as they might eventually be found within the FLONA Sincorá, but are not featured otherwise in this paper.

1. *Arrojadoa* Britton & Rose

1.1. *Arrojadoa penicillata* (Gürke) Britton & Rose, *Cact*. **2**: 171 (1920).
Common name: rabo-de-raposa.
Fig. 2A.

Erect to decumbent, sparsely branched shrub, normally reaching 1m tall but sometimes when growing amongst other shrubs up to 3.5m tall. Branches cylindric, 1–1.8cm diam., thicker below the cephalium, with a strongly lignified vascular cylinder; ribs 8–12, 2–4×3–7mm; epidermis olive-green; areoles 2mm diam., 5–9mm apart, spines reddish brown, delicate, central spines 2–4, 5–10mm, ± deflexed, radial spines 8–10, 2–7mm. Floriferous region forming a ring cephalium terminating the year's growth with abundant white wool and reddish brown bristles to 1.5–2.5cm. Flowers 1.5–3×2cm, uniformly magenta to reddish pink, opening in the late afternoon; pericarpel naked, 0.5–1.0×0.3–0.5cm; floral tube 1.2×0.4–0.7cm; perianth-segments triangular to lanceolate, c. 0.7×0.3cm, outer segments spreading to patent, inner segments erect; stamens included in relation to the perianth-segments; style 10mm long, stigma-lobes 6–8, 1mm long, at the same height as the stamens; ovary obtriangular in longitudinal section. Fruit indehiscent, 0.9cm long, obovoid, pericarp smooth, greenish to dark red, funicular pulp white. Seed c. 1.2mm long.

Arrojadoa penicillata is a Brazilian endemic cactus from the states of Bahia and Minas Gerais (Zappi *et al*., 2015). It has been observed that native bees and birds shelter in and around the plants, and that the fruits serve as food for native animals in the caatinga (Silva, 2007), while the flowers are visited by hummingbirds. The native habitat of the species is quite extensive, however, it is subject to widespread degradation caused by grazing by cattle and goats, and also by urban development and occasional fires. Within the FLONA Sincorá the species occurs at altitudes between 200 and 405m, in caatinga *sensu stricto* with rock outcrops, and also on the roadside, with c. 20 individuals along the road, with fragmented distribution and, unfortunately, cattle roam freely within the FLONA Sincorá. This species was found flowering and fruiting between January and April.

Outside the study area, at the Serra do Voador (13°54'57.0"S, 41°04'04.0"W), we observed populations of *Arrojadoa rhodantha* (Gürke) Britton & Rose that differs from *A. penicillata* in having more ribs (10–14), stouter stems (1.8–6.0cm diam.) that are not widened towards the ring cephalium, as well as more robust spination.

Identification key

1. Leaves with expanded blade; stems not succulent and not obviously photosynthetic
  2. Flowers yellow; fruits globose, c. 4-seeded, to 1.3mm diam., seeds brownish 4–8 × 5–7mm .....8.1. ***Pereskia aureiflora***
  2'. Flowers pink; fruits pear-shaped, 20–30-seeded, to 5.5mm diam., seeds dark brown to black, 1–2,5 × 2mm .....8.2. ***Pereskia bahiensis***
1'. Leaves absent or reduced, turbinate, present only in young, growing branches; stems succulent, photosynthetic
  3. Areoles with glochids or microscopically serrated spines; ribs absent; distal branches often flattened and rounded and with determinate growth; seed testa enveloped by pale brown fibres
    4. Trees with dimorphic branches; main stem cylindric and lateral terminal stems flattened, obovate; perianth-segments bright to green-yellow, seeds 8–10mm diam. .....6.1. ***Brasiliopuntia brasiliensis***
    4'. Shrubs to subshrubs, branches cylindric or flattened, not dimorphic; perianth-segments red to orange or purple-green; seeds 4–5mm diam.
      5. Branches cylindric; outer perianth-segments purplish-green.....11.2. ***Tacinga funalis***
      5'. Branches flattened, perianth-segments red to orange-red.
        6. Areoles sunken, punctiform, spineless; fruits globose.....11.1. ***Tacinga inamoena***
        6'. Areoles prominent, with sharp golden or grey spines; fruits ovoid to subglobose
          7. Fruit subglobose, green to purple or reddish when ripe, funicular pulp green..... .....11.3. ***Tacinga palmadora***
          7'. Fruit ovoid, white when ripe, funicular pulp pink.....*Tacinga werneri*
  3. Areoles without glochids or microscopically serrated spines; branches cylindric, ribs present; seed testa naked, dark brown or black
    8. Climbers; branches trigonous, areoles star-shaped, with up to 6 short spines reaching 0.5cm .....6.1. ***Hylocereus setaceus***
    8'. Shrubs to solitary-stemmed plants; stem with distinct ribs, areoles with more than 6 spines generally longer than 0.5cm
      9. Plants unbranched, globose to cylindric reaching 25cm tall; with a terminal cephalium
        10. Globose plants, less than 13cm tall, ribs 8–12, flowers 0.6–1.0cm wide 7.1. ***Melocactus concinnus***
        10'. Cylindric plants (9.5–)13–21cm tall, ribs 8–14, flowers 1–1.2cm wide .....*Melocactus inconcinnus*
      9'. Branched shrubs or trees, branches columnar, flowering region not differentiated or flowers in a lateral or ring-like cephalium
        11. Floriferous region differentiated into a lateral or ring-like cephalium covering 3 or more ribs at the apex of the plants, or ring-like
          12. Branches with a lateral cephalium covering 3 or more ribs, fruits totally immersed in wool .....5.1. ***Espostoopsis dybowskii***
          12. Branches with an apical ring cephalium marking the annual growth of the branches, fruits not immersed in wool
            13. Stems less than 2cm diam., spines reddish; flowers 1,5–3cm long, perianth-segments magenta to pink-red
              14. Branches c. 1.8cm diam., wider below the cephalium, ribs 7–12, central spines 2–4, 0.2–0.5cm; perianth-segments expanded, flower c. 2cm wide.....1.1. ***Arrojadoa penicillata***
              14'. Branches 2–6cm diam., not thickened below the cephalium, ribs 8–14, central spines 4–5, 1.5–4cm, perianth-segments erect, 8–10cm long .....*Arrojadoa rhodantha*
            13'. Stems over 3cm diam., spines yellow to white; flowers 5–7cm long, perianth-segments green to white.....10.1. ***Stephanocereus leucostele***
        11'. Flowering region not differentiated
          15. Fruit indehiscent, flower-tube and pericarpel covered in wide, reddish green scales .....2.1. ***Brasilicereus pheacanthus***
          15'. Fruits dehiscent by a longitudinal or irregular slit, pericarpel smooth, not scaly
            16. Flowering areoles glabrous; flowers over 17cm long; fruit ovoid, red or pink when ripe, funicular pulp white .....4.1. ***Cereus jamacaru***
            16'. Flowering areoles hairy; flowers up to 7cm long, fruit globose to depressed-globose, purplish to wine coloured when ripe, funicular pulp magenta.
              17. Shrubs without a well-defined trunk with candelabra-like, decumbent branches, fruit globose .....*Pilosocereus gounellei* subsp. *gounellei*
              17'. Tree-like, branches semi-erect, irregularly branched, fruit depressed-globose
                18. Ribs (3–)4–6, epidermis grey to olive-green, areoles hairy ....9.1. ***Pilosocereus catingicola*** subsp. ***catingicola***
                18'. Ribs (5–)6–10, epidermis blue, areoles glabrous..... .....*Pilosocereus pentaedrophorus* subsp. *robustus*

Examined material: Bahia, Mun. Contendas do Sincorá, Floresta Nacional Contendas do Sincorá, trilha das Bromélias, 13°45'46.1"S 41°02'28.0"W, 12. May 2012 *V.V.F. Lima et al.* 15 (CEN); l.c., 13°54'57.0"S 41°04'04.0"W, 4 Apr. 2014, *L.Y.S. Aona et al.* 3552 (HURB); l.c., ramal contrário à entrada para a sede da Floresta Nacional Contendas do Sincorá, 13°55'46"S 41°05' 16" W, 6 Feb. 2015, *L.Y.S. Aona et al.* 3962 (fl.) (HURB); l.c., Trilha via Rio Combuca, próximo a sede da Floresta Nacional Contendas do Sincorá, 13°55'05" S 41°06'53" W, 6 Feb. 2015, *L.Y.S. Aona et al.* 3959 (fl.) (HURB).

2. *Brasilicereus* Backeb.

2.1. *Brasilicereus phaeacanthus* (Gürke) Backeb., *Jahrb. Deutsch. Kakteen-Ges.* 1941(**2**): 50 (1942).

Erect, sometimes semi-decumbent, branched shrub, normally around 3m tall, but, when growing amongst other shrubs, up to 5m tall. Branches columnar, 2.5–6cm diam., cylindric, with strongly lignified vascular cylinder; 8–12 ribs, 5×6mm; epidermis dull to grey-green; areoles never woolly, 2–3mm diam., 5–10mm apart, spines white to yellowish, central spines 1–3, 5–25mm, erect, radial 10–14, 3–10mm. Floriferous region not differentiated. Flowers 4–7×5cm, externally greenish to reddish, opening in the early evening and with strong odour; pericarpel 0.5–1×1.5–1.8cm, covered by green scales; floral-tube infundibuliform, 3.2×1.7–2cm, covered in reddish-green or slightly purplish bracts; perianth segments triangular to obovate, 1–1.4×2.5cm, outer segments spreading to patent, internal spreading to erect, white; stamens included in relation to the perianth-segments; style 20–25mm long, stigma-lobes 10, presented above the stamens; ovary depressed-obtriangular in longitudinal section. Fruit indehiscent, globose to ovoid, 3.5cm diam., pericarp scaly, greenish to dark red or slightly purplish, funicular pulp white. Seed 2–3mm long.

Endemic from eastern Brazil, this species occurs in the states of Bahia and Minas Gerais in areas of the Caatinga biome bordering the Atlantic Forest, in a vegetation type known as 'agreste', often associated with granite/gneissic outcrops (Taylor & Zappi, 2004). The flowers are possibly pollinated by bats and the fruits are sought after by birds, especially small parrots. With a fragmented distribution, this species is subject to habitat degradation by cattle farming and agricultural and mining developments, besides an increasing risk of fire, all this contributing to its shrinking distribution. Within the FLONA Sincorá it is found in tall, well preserved caatinga, in four or five populations each with a few (5–8) individuals. Only sterile specimens were observed at the FLONA Sincorá during this project.

Examined material: Bahia, Mun. Contendas do Sincorá, Floresta Nacional Contendas do Sincorá (FLONA), 13°55'07.6"S, 041°06'58.2"W, 2 Apr. 2014, *M.B. Medeiros et al.* 601 (ALCB, HURB, R); l.c., Trilha via Rio Combuca, próximo à sede da Floresta Nacional Contendas do Sincorá, 13°55'05"S, 41°06'53" W, 6 Feb. 2015, *L.Y.S. Aona et al.* 3961 (HURB).

Additional material: Mun. Boa Nova, Parque Nacional de Boa Nova; Setor oeste, Fazenda Lagedão, 14°24'46.0"S 40°07'46.0"W, 3 Mar. 2014, *L.Y.S. Aona et al.* 2225 (fl., fr.) (HURB); Mun. Tanhaçu, 21km ao N Sussuarana, estrada para Contendas do Sincorá, 3 Feb. 1991, *N.P. Taylor et al.* 1547 (CEPEC, HRCB, K, ZSS).

3. *Brasiliopuntia* A. Berger.

3.1. *Brasiliopuntia brasiliensis* (Willd.) A. Berger., *Entwickl. Kakteen*: 17, 18, 94 (1926).
Common names: urumbeba, facho-de-renda.

Trees reaching over 10m tall, with a cylindric trunk covered in clusters of spines to 9cm long, reaching 35cm diam. in older specimens. Branches dimorphic, patent in relation to the trunk, whorled, with cylindric intermediate joints 15–20(–100)cm long and ultimate joints flattened, obovate, 3–8×1.5–3cm, epidermis dark-green, shiny; areoles sparsely spread, 15–30mm apart, white felted, with or without glochids, with 0–1 spines in the determinate growth joints, 1–3 spines in the indeterminate sections, spines grey to white. Rudimentary leaves borne on young, growing branches, ovoid to cone-shaped, succulent, bright green, soon deciduous. Floriferous region not differentiated, flowers appearing from the edge of the determinate growth joints or from near the apex of the leader shoot. Flowers 2.5–3.5×4.5cm, opening during the day and without noticeable scent; pericarpel hemi-globose, 2–2.8×0.9–1.2cm, pale green, with unarmed areoles with abundant glochids; perianth-segments yellow-green to bright yellow, outer perianth-segments 1.5cm long, ovate, patent; inner segments 2×1cm, lanceolate to spathulate, spreading to patent; stamens 7mm long, incurved, filaments not sensitive, anthers 0.6mm, white; style 9mm long, stigma-lobes 3–6, presented above the stamens; ovary rounded in longitudinal section. Fruit

indehiscent, globose to ovoid, sometimes pear-shaped, 2–4cm diam., pericarp purple to red or orange-yellow, with areoles covered in glochids, funicular pulp fibrous. Seed 8–10mm long, 1–5 per fruit, aril fibrous, pale brown.

Distinct from the other genera of Opuntioideae, the dimorphic branches found in the genus *Brasiliopuntia* resemble the Caribbean *Consolea*. Formerly accepted as a monotypic genus (Taylor & Zappi, 2004; Hunt *et al*., 2006), it has recently been expanded to include a second species from Argentina (*B. schickendantzii* (F.A.C. Weber) R. Puente & Majure).

It occurs in most extra-Amazonian states of Brazil, excepting Rio Grande do Norte, Piauí, Goiás and the Distrito Federal, where it has not yet been collected. Outside Brazil, its distribution extends into dry and seasonal forests in Argentina, Bolivia, Peru and Paraguay. In Brazil it occurs in many vegetation-types, often associated with rocks and sandy soils (Taylor & Zappi, 2004). It has been introduced to Florida, USA.

With a wide altitudinal tolerance, it occurs from sea-level to 1,000m. At the FLONA Sincorá a single individual was located at 366m alt., in well preserved caatinga.

Examined material: Floresta Nacional Contendas do Sincorá 13°55'08.0"S 41°06'52.0"W, 3 Apr. 2015, *M.B. Medeiros et al.* 603 (HURB) l.c., 50m da sede, via Rio Combuca, 13°55'18"S, 41º07'08"W, 6 Feb. 2015, *L.Y.S. Aona et al.* 3847 (HURB).

Additional material: São Paulo, Mun. Atibaia, Dec. 2001, *L.Y.S. Aona et al.* 776 (fl., fr.) (UEC).

4. *Cereus* Mill.

4.1. *Cereus jamacaru* DC., *Prodr*. **3**: 467 (1828). Common names: mandacaru, mandacaru-da-Bahia.
Fig. 2B.

Trees reaching over 12m tall, with a cylindric, often short but well defined trunk covered by clusters of stout spines often longer than 10cm in older specimens. Branches columnar, 4–6 ribbed, 7–20cm diam., constricted in regular intervals, erect to spreading in relation to the trunk, epidermis grey-green to glaucous, waxy; areoles 3–5mm diam., 15–20mm apart, slightly woolly, spines slender and golden when young, turning grey with black tips when old, spination very variable amongst individuals, central spines 1–5, 2.5–15cm long, radial spines 8–12, 1–3.5cm long. Floriferous region not differentiated, flowers appearing mostly from the ribs of upper branches. Flowers 17–30×10–20cm, opening at dusk and remaining open until early morning of the next day, fragrant; pericarpel hemi-globose to shortly cylindric, 2–2.8×1.2–1.8cm, green, unarmed, sparsely scaly and slightly ribbed; flower-tube infundibuliform, 12–14×5cm at apex, pale green with sparse scales, outer perianth-segments 5–8×2–2.2cm long, pale green, sometimes reddish at tips, elongate spathulate; inner segments 1–2×0.6–0.9cm, pure white, elongate spathulate, sometimes frilly and acuminate to mucronate at apex, spreading to reflexed; stamens numerous, delicate, presented just below the stigma-lobes, anthers globose; style 50–170mm long, white to greenish, stigma-lobes 12–16, 11–19mm long, spreading above the stamens; ovary hemi-circular to rectangular in longitudinal section. Fruit dehiscent by a longitudinal slit, egg-shaped, 3–8×1.5–3cm diam., pericarp deep red to pinkish, funicular pulp white. Seed 1–2 ×1mm, numerous, blackish, shiny.

A Brazilian endemic found in all states of northeastern Brazil (Zappi *et al*., 2015), occuring at altitudes from 50 to 1,200m (Taylor & Zappi, 2004), in a wide variety of habitats, from coastal to montane, but mostly distributed in the Caatinga biome (Taylor & Zappi, 2004, Menezes *et al*., 2011). Widely cultivated and abundant along roadsides and in degraded caatinga, it is often preserved by farmers as it can be used as cattle-fodder during droughts, and also because of its edible fruit. It is also used to build living fences and, in the past, its wood was used for building houses locally. Despite not being endangered as a species, large individuals that could be used for logging are extremely rare nowadays.

Within the FLONA Sincorá this species occurs together with *Pilosocereus catingicola* subsp. *catingicola*, in caatinga with trees and shrubs and rock outcrops, at altitudes around 300–440m alt., in a population with 10–15 specimens. Isolated individuals are often found by the access roads. It flowers and fruits between January and April. Cattle grazing and evidence of recent fire were noticed around the area.

Examined material: Bahia, Mun. Contendas do Sincorá, Floresta Nacional Contendas do Sincorá, 50m da sede, via Rio Combuca, 13°55'19"S, 41°06'57"W, 6 Feb. 2015, *L.Y.S. Aona et al*. 3856 (fl., fr) (HURB).



Figure 2. Cactaceae at the FLONA Sincorá, Bahia, Brazil. A. *Arrojadoa penicillata*, habit. B. *Cereus jamacaru*, habit. C–D. *Espostoopsis dybowskii*, C. habit; D. branch. E–F. *Melocactus concinnus*. E. habit; F. Cephalium with open flowers. G–H. *Pereskia aureiflora*. G. flower; H. Fruit

G–H: photographs by Lucas Marinho.

5. *Espostoopsis* F. Buxbaum

5.1 *Espostoopsis dybowskii* (Rol.-Goss.) Buxb. in KRAINZ, *Die Kakteen*, Lfg **38–39** (1968).
Common names: homem-velho, mandacaru-cabeça-branca, mandacaru-de-penacho.
Figs. 2C–D.

Shrubs 2–3m tall (reaching 5m in N Bahia), branched only at base (sometimes at apex when cut or broken). Branches columnar, 20–26-ribbed, 4–8cm diam., without constrictions, erect, epidermis grey-green to glaucous, waxy; areoles 2–3mm., 6–15mm apart, slightly woolly, spines slender and golden when young, turning grey with black tips when old, spination very variable amongst individuals, central spines 1–2(3) 0.5–2cm long, radial spines 10–14, 0,3–1cm long. Floriferous region profoundly differentiated, appearing mostly on upper part of the branches and totally covering 3–6 ribs with dense wool and golden bristles. Flowers deeply inserted in the cephalium, 1.7–3×1–2cm, opening at dusk and remaining open until early morning of the next day; pericarpel hemi-globose to shortly cylindric, 2–2.8×1.2–1.8cm, green, unarmed, sparsely scaly and slightly ribbed; flower-tube infundibuliform, 12–14×5cm at apex, pale-green with sparse bract-scales, outer perianth segments 5–8×2–2.2cm, pale green, sometimes reddish at tips, elongate spathulate; inner segments 0.6–0.9×0.2–0.3cm, pure white, elongate spathulate, sometimes frilly and acuminate to mucronate at apex, spreading to reflexed; stamens numerous, delicate, presented just below the stigma-lobes, anthers globose; style 50–70mm long, white to greenish, stigma-lobes 12–16, 11–19mm long, spreading above the stamens; ovary hemicircular to rectangular in longitudinal section. Fruit dehiscent by a longitudinal slit, egg-shaped, 5.5–8.5×2.6–3.5cm diam., pericarp deep red to pinkish, funicular pulp white. Seed 1–2×1mm, numerous, blackish, shiny.

Endemic to Bahia, *Espostoopsis dybowskii* has two major disjunct localities, a very isolated one near Juazeiro, and a group of southern populations in the region of the Rio de Contas and Contendas do Sincorá (Taylor & Zappi, 2004). Apart from the subpopulations recorded here, there is another further south, at the Serra Escura near Sussuarana, in the municipality of Tanhaçu (N.P. Taylor pers. comm.) At the FLONA Sincorá the species is found at 400–700m alt. on gneissic rock outcrops, with c. 40 individuals. A further population with c. 40 individuals was recorded at the Serra do Voador, outside the limits of the FLONA Sincorá, growing in the proximity of *Melocactus inconcinnus* (Figure 4). This last area is under pressure from local fires, cattle grazing and wood extraction, and the health of this cactus population is already compromised.

Habitat loss for the species is due to its transformation for agriculture and cattle farming affecting both southern populations. The studied population is also at risk from fire and stone-quarrying (Taylor & Zappi, 2004), and the species is considered Vulnerable under the IUCN criteria (IUCN, 2015), and Taylor & Zappi (2004) had suggested that at least part of its range should be included in a preserved area. The discovery of around 50 individuals within the area protected by the FLONA Sincorá is thus very timely, however, there is evidence of fire-damaged specimens, corroborating the threats known to be affecting this species. *E. dybowskii* was collected with floral remnants in February but mature fruits were not observed.

Examined material: Bahia, Mun. Contendas do Sincorá, Floresta Nacional Contendas do Sincorá, Serra da Volta, 13°46'45"S, 41°03'09" W, 23 Jul. 2014, *M.R. Peixoto et al.* 6 (HURB); l.c., Serra da Volta, 13°46'54"S, 41°02'56"W, 7 Feb. 2015, *L.Y.S. Aona et al.* 3927 (fl., fr.) (HURB).

Additional material: Mun. Maracás, 7km E of Porto Alegre on the road to Jequié, *N.P. Taylor et al.* 1551 (CEPEC, HRCB, K, ZSS).

6. *Hylocereus* (A. Berger) Britton & Rose

6.1. *Hylocereus setaceus* (Salm-Dyck) R. Bauer, in *Cact. Syst. Initiatives* **17**: 29 (2003).
Common names: rainha-da-noite, mandacaru-de-três-quinas.

Hemi-epiphytic to epiphytic plants, often germinating on the ground and climbing on rocks or tree-trunks. Branches trigonous, 0.5–1.5cm wide, bright green, irregularly branched; areoles spiny, 0.2–0.5cm diam., 1.5–2cm apart, spines conic, hard-tipped, centrals 1–3, 0.2–0.5cm, radials 4–6, 0.1–0.4cm; young areoles with golden bristles, floriferous areoles glabrous, not differenciated. Flowers solitary, lateral or subapical, 20–32×22–25cm; pericarpel green, covered in spiny areoles; outer perianth-segments 9–10cm long, linear-lanceolate, reddish green, fleshy, reflexed, inner segments to 12cm long, lanceolate, white, delicate, erect. Fruit ovoid, perianth remnants deciduous, indehiscent, pericarpel reddish green, initially covered in spiny areoles that fall when the fruit ripens, funicular pulp white. Seeds sub-orbicular, 2.5mm long.

Hylocereus setaceus has characteristic trigonous stems and large flowers with a spiny pericarpel that distinguish it from all other cactus species found in the FLONA Sincorá. The commonly cultivated, sometimes naturalized *Hylocereus undatus* differs by having crenate, thickened rib margins and a pericarpel and tube covered in wide scale-bracts (vs straight, not thickened, scale-bracts absent in *H. setaceus*) (Taylor & Zappi, 2004).

Widespread in South America, this species occurs in a variety of forests and rock outcrops in the majority of the Brazilian biomes and in six states in northeastern Brazil (Taylor & Zappi, 2004). Large tracts of one of its natural habitats, the Atlantic Rainforest, have largely been destroyed, thus the species' extent of occurrence has greatly diminished. However, it occurs within several protected areas and is found in many locations, therefore it is not considered threatened. At the FLONA Sincorá only two sterile individuals were observed growing as epiphytes in tall caatinga forest.

Examined material: Bahia, Mun. Contendas do Sincorá, Floresta Nacional Contendas do Sincorá, (FLONA), 13°54'57.1"S, 41°04'4.5"W, 4 Apr. 2014, *M.B. Medeiros et al.* 605 (HURB).

Additional material: ESPÍRITO SANTO, Afonso Cláudio, estrada para Itarana, 570m alt., 24 Nov. 1999, 20º03'08"S, 41º03'03"W, *D. Zappi et al.* 411 (K, UEC).

7. *Melocactus* (L.) Link & Otto

7.1. *Melocactus concinnus* Buining & Brederoo, in *Kakt. and. Sukk.* **23**: 5–7 (1972).
Common name: cabeça-de-frade.
Fig. 2E–F.

Globose, solitary, unbranched plants to 8–10×10–14cm, internally mucilaginous, epidermis green to grey-green, glaucous when young; ribs 10–12, (30–)20×20–40mm; areoles 5–6×6mm, 1.5–3cm apart; young spines red to black, turning pale yellow to brownish and finally grey with age; central spine (0–)1, 10–15mm long, curved downwards; radial spines 6–7, slightly recurved, 1–3 longer than the others, 15–25mm long. Terminal cephalium well-developed, to 5×6–9 cm, bristles pale red to pale rose-pink. Flowers radial, deeply inserted in the cephalium, 2–2.2×0.6–1cm; flower-tube c. 0.6–1cm long; pericarpel and flower-tube unarmed, devoided of scales; perianth-segments linear, pale pink, outer segments reflexed, inner segments 0.6cm long, erect; style 12–13mm, stigma-lobes 3–5, c. 1mm long. Fruit clavate to turbinate, 12–13×2–3mm, indehiscent, pinkish, pushed out of the cephalium when ripe; seeds not observed.

Endemic from central parts of eastern Brazil, *Melocactus concinnus* occurs in central/southern Bahia and northeastern Minas Gerais, in caatinga, cerrado and ecotonal vegetation in the campo rupestre highlands, on stony soil with quartzitic sand and/or gravel, or in crystalline rock crevices at 550–1,300m alt. (Taylor & Zappi, 2004, Zappi *et al*., 2015). The caatinga vegetation has undergone transformation into agricultural land, exploitation for fuel-wood and fires, however, the campo rupestre vegetation is still in a reasonably good state of conservation and the populations of this species are abundant, thus not being considered threatened (IUCN, 2015). Other pressure faced by species of *Melocactus* is that they are often collected for local use as medicinal and ornamental plants, and sold by the roadside, and sometimes they may be targeted by plant collectors (Taylor & Zappi, 2004).

M. concinnus was collected by the road at the FLONA Sincorá, where its populations are vulnerable to habitat disturbance (Figure 4). A small population with around 15 individuals was found along the track that cuts across this protected area. It was found flowering in February and April.

Examined material: Bahia, Mun. Contendas do Sincorá, Floresta Nacional Contendas do Sincorá (FLONA), ramal contrário a entrada da Sede da Floresta Nacional Contendas do Sincorá, 13°57'11"S, 41°02'25"W, 6 Feb. 2015, *L.Y.S. Aona et al.* 3900 (HURB); l.c., Lado direito da BR-026, em direção à fazenda Vargem Grande, Contendas do Sincorá, 13°45'46.1"S, 41°02'28.0"W, 10 May 2012, *V.V.F. Lima et al.* 4 (CEN); l.c., Fazenda Pé de Morro, próximo ao Rio Medalha, Contendas do Sincorá, 13°45'46.1"S, 41°02'28.0"W, 12 May 2012, *V.V.F. Lima et al.* 13 (CEN).

8. *Pereskia* Mill.

8.1. *Pereskia aureiflora* F. Ritter, *Kakt. Südamer.* **1**: 22 (1979).
Common names: facho, ora-pro-nobis-da-mata.
Fig. 2G.

Shrub or tree to 6m alt., trunk up to 20cm diam. Branches with areoles 2–3mm diam., 1.5–3cm apart, spines 1–3, straight, erect., 1.5–2cm. Petiole 2–4mm long; leaves elliptic to lanceolate, 3–5.5×2.5–3.5cm, base broadly cuneate, apex acute, lateral veins 3–5(–7). Flowers solitary, terminal or on lateral branches, yellow, 3–4cm diam.;

pedicel short, pericarpel obovoid with areoles and bracts; outer perianth-segments 3–4, obovate, yellow; inner segments 0.7–1.8×0.5–1.8 cm, obovate, bright yellow; stamens 5–10mm long, anthers cream-yellow; ovary semi-inferior, style 10mm long, white, stigma-lobes 4–7, erect. Fruit globose, to 1.3cm diam., 1–1.3×0.8–1.2cm, pericarp green to brownish, with 5–8 patent bracts on the lower half and 5–8 erect bracts on the upper half. Seeds c. 4, 4–8×5–7 mm, obovoid, shiny, brownish, laterally compressed, testa smooth.

Endemic to eastern Brazil, *P. aureiflora* occurs in Bahia and Minas Gerais, where it grows associated with gneissic/granite inselbergs at elevations of 170 to 920m (Taylor & Zappi, 2004, Zappi *et al*., 2015), and can be easily distinguished by its golden yellow flowers, however it is easily overlooked when sterile.

The extent of occupancy of this species has diminished due to degradation and deforestation of the caatinga for agriculture, as well as granite and gneiss rock extraction, thus it is listed as Endangered (IUCN, 2015). At the FLONA Sincorá the species is found in well preserved tall caatinga near the Rio Combuca (Figure 4), in a small population (c.7 individuals). There was earlier evidence of another population near the track that cuts through the protected area, but no individuals were found there recently. It was collected in flower between February and April.

Examined material: Bahia, Mun. Contendas do Sincorá, Floresta Nacional Contendas do Sincorá, 50m da sede, Rio Combuca, 13°55'05"S, 41°06'53"W, 6 Feb. 2015, *L.Y.S. Aona et al.* 3848 (fr.) (HURB); l.c., trilha do Rio Cumbuca, 14°01'17.0"S, 41°14'53.2"W, 26 Feb. 2012, *L.C. Marinho* 281 (HUEFS);

8.2. *Pereskia bahiensis* Gürke, in *Monatsschr. Kakt.-Kunde* **18**: 86 (1908)..
Common name: Quiabento, flor-de-cera.
Fig. 3A

Shrub or treelet 3–4.5m alt., eventually forming a trunk 15–20cm diam. Branches with areoles 5–7mm diam., 1.5–3cm apart, spines 1–5, straight, erect., 1.5–4cm. Petiole short to absent; leaves elliptic to obovate, 6–9×4.5–7cm, base cuneate, apex obtuse, sometimes emarginate, lateral veins indistinguishable. Flowers in terminal cymes, pink, 4–7cm diam.; pedicel 1cm, pericarpel obcuneate in outline, with areoles and bracts; outer perianth-segments 2–5, obovate, pinkish; inner segments 0.7–0.9×2–3 cm, obovate to spathulate, pink to deep magenta; stamens 5–10mm long, anthers cream-yellow; ovary inferior, style 7–15mm long, white, stigma-lobes 4–7, erect. Fruit pear-shaped, to 5.5cm diam., pericarp green turning yellow when ripe (when fallen to the ground), bracts deciduous. Seeds 20–30, obovoid, shiny, dark brown to black, 1–2.5×2 mm, laterally compressed, testa smooth.

Endemic to the state of Bahia, *P. bahiensis* occurs in the caatingas surrounding the Chapada Diamantina, Maracás Plateau, northern Serra Geral, to the east and south of the Rio São Francisco, between 300 and 900m altitude (Taylor & Zappi, 2004). Common in the Serra do Sincorá region, it is often used as a living fence. At the FLONA Sincorá it is relatively abundant in the shrubby and tall caatinga and surrounding rock outcrops where there was evidence of fire and cattle grazing. It was observed flowering and fruiting between January and April.

Examined material: Bahia, Mun. Contendas do Sincorá, Floresta Nacional Contendas do Sincorá (FLONA), 13°54'56.8"S, 041°04'04.2"W, 4 Apr. 2014, *L.Y.S Aona et al*. 3549 (HURB); l.c., 50m da sede, Rio Combuca, 13°55'18"S, 41°07'08"W, 6 Feb. 2015, *L.Y.S. Aona et al.* 3854 (fr.) (HURB); l.c., Serra da Volta, 13°46'54"S, 41°02'56"W, 7 Feb. 2015, *L.Y.S. Aona et al.* 3925 (HURB); l.c., próximo à sede, 3 Sep. 2011, *L.C. Marinho* 86 (HUEFS).

9. *Pilosocereus* Byles & G.D. Rowley

9.1. *Pilosocereus catingicola* (Gürke) Byles & G.D. Rowley in *Cact. Succ. J. Gr. Brit.* **19**: 66 (1957) subsp. ***catingicola***.
Common names: facheiro, mandacaru-babão, mandacaru-de-facho.
Figs. 3B–D.

Tree-like plants to 6–7m tall, with a well-defined trunk. Branches columnar, erect, vascular cylinder strongly lignified, 4–5-ribbed, 7–14cm diam., ribs 30–35×15–30mm, without constrictions, epidermis olive-green to glaucous, waxy; areoles 5–10mm., 10–12mm apart, white woolly at apex, wool brown in old areoles; spines slender and straw-coloured when young, turning grey when old, central spines 1–3, 1.5–3cm long, radial spines 7–10, 0.5–1.5cm long. Floriferous region weakly differentiated, flowering areoles sparse, apical to subapical, with white to grey wool. Flowers 5–7×3.5 cm, opening at dusk and remaining open until early morning of the next day; pericarpel hemi-globose, green, unarmed; flower-tube cylindric, 3.5–4cm long, green, outer perianth-segments 10–13, 1.7×2.5cm, elongate spathulate, greenish with red edges, inner segments 16–18,



Figure 3. Cactaceae at the FLONA Sincorá, Bahia, Brazil. A. *Pereskia bahiensis*, flower. B–D. *Pilosocereus catingicola* subsp. *catingicola*. B. habit; C. old flower and flower-bud; D. fruit. E–F. *Stephanocereus leucostele*. E. open flower; F. fruiting branch. G. *Tacinga inamoena*, fruiting branch. H. *Tacinga palmadora*, fruiting branch.

3–3.5cm, white with pinkish streaks; stamens numerous, anthers globose; style 3.5–4cm long, pale yellow, stigma-lobes 12, 5–6mm long, spreading above the stamens. Fruit depressed-globose, dehiscent by central irregular slit, 4×3.5cm, perianth remnants remaining attached, pericarp wine-coloured to purplish when ripe, funicular pulp magenta. Seeds cochleariform 1–2×2mm, numerous, blackish, shiny.

Pilosocereus catingicola is endemic to north-eastern Brazil, occurring in the states of Alagoas, Bahia, Ceará, Paraíba, Pernambuco, Rio Grande do Norte and Sergipe, in variable substrates with the typical subspecies, subsp. *catingicola* occurring in the caatinga of Bahia (Taylor & Zappi, 2004, Zappi *et al*., 2015). This subspecies is abundant in eastern Bahia and, despite having lost considerable area of its habitat because of degradation and deforestation of the caatinga, it occurs in a few protected areas and the remaining populations are still thriving, thus it is not currently considered endangered.

At the FLONA Sincorá it was found at 200–400m alt. in shrubby caatinga, both inside and near the protected area, sometimes growing in pasture and by roads, with a fragmented distribution along the track that cuts through the protected area; around 15 individuals were counted. Throughout the region, local farmers often leave the plants when clearing the fields for cattle grazing, as their fruits and branches can be used forage in dry periods. It was found flowering in March.

Examined material: Bahia, Mun. Contendas do Sincorá, Floresta Nacional Contendas do Sincorá, (FLONA), Serra da Volta, 13°46'45"S, 41°03'09"W, 23 July 2014, *M.R. Peixoto et al.* 4 (HURB); l.c., ramal contrário à entrada para a Sede da Floresta Nacional Contendas do Sincorá, 13°55'49"S, 41°05'16"W, 6 Feb. 2015, *L.Y.S. Aona et al.* 3888 (fr., fl) (HURB); l.c., 24. Mar. 2013, *M.O.T. Menezes* 312 (EAC).

10. *Stephanocereus* A. Berger

10.1. *Stephanocereus leucostele* (Gürke) A. Berger, *Entwicklungslin. Kakt.*: 59, 97 (1926). Figs. 3E–F.

Little branched, tall shrubs or small trees with decumbent columnar branches constricted by the formation of a cephalium at the end of each growing season. Branches columnar, erect, vascular cylinder strongly lignified, 16–18-ribbed, 4–10(–15)cm diam., ribs 7–8×5–6mm, epidermis grey-green, waxy; areoles 3–5mm., 6–10mm apart, with abundant white wool that covers the stem, spines slender, white or golden, central spines 1–2(–3), 1–2.5cm long, radial spines 14–18, 0.6–1.3cm long. Floriferous region forming a ring cephalium with white wool and fine, golden bristles. Flowers 5–7×4–5.3cm, opening at dusk and remaining open until early morning of the next day, with strong cabbage odour; pericarpel hemiglobose, green, unarmed; flower-tube cylindric, slightly curved, 5–6×2.2–3.5cm, pale green, smooth, outer perianth-segments 15–18cm long, lanceolate, greenish with red apex, inner segments pure white; stamens numerous, anthers globose; style 60mm long, pale yellow, stigma-lobes 7–8, 1–2mm long, spreading above the stamens. Fruit globose, narrowed at the base, indehiscent, 3.5–4×3.5cm, perianth remnants remaining attached, pericarp dark green to dark blue, funicular pulp white or red. Seeds cochleariform 2–2.2mm long, numerous, blackish, dull.

Endemic to the caatinga of southern and central Bahia that surrounds the Chapada Diamantina, this is one of the most characteristic cacti found in the region were its distribution is associated with the areas surrounding the highland (Taylor & Zappi, 2004). It is easily identified by its ring-cephalium and white-woolly stems. Despite the fact that its habitat is under pressure through destruction of the caatinga for fuel-wood and agriculture, its populations are widely distributed and found in protected areas, such as the FLONA Sincorá and Gruta dos Brejões (Taylor & Zappi, 2004).

In the FLONA Sincorá the species was collected at 300–700m alt. in shrubby caatinga, with larger numbers of individuals (c. 20) growing on the higher ground, in very rocky soil. Flowers and fruits were observed between January and April.

Examined material: Bahia, Mun. Contendas do Sincorá, Floresta Nacional Contendas do Sincorá, (FLONA), 13°55'08.6"S, 041°06'52.3"W, 2 Apr. 2014, *B.M. Medeiros et al*. 602 (HURB); l.c., ramal contrário à entrada para Sede da Floresta Nacional Contendas do Sincorá, 13°55'46"S, 41°05'16"W, 6 Feb. 2015, *L.Y.S.Aona et al*. 3966 (HURB); l.c., 13°55'08.6"S, 41°06'52.3"W, 2 Apr. 2014, *B.M. Medeiros et al.* 602 (R); l.c., estrada que da acesso ao portão principal, 13°45'46.1"S, 41°02'28.0"W, 14 May 2012, *V.V.F.Lima et al.* 16 (CEN); l.c., Lado direito da BR-026, seguindo pela estrada de terra, em direção a Vargem Grande, 13°45'46.1"S, 41°02'28.0"W, 11 May 2012, *V.V.F.Lima et al.* 5 (CEN); l.c., 2 Apr. 2014, *B.M. Medeiros et al*. 602 (ALCB).

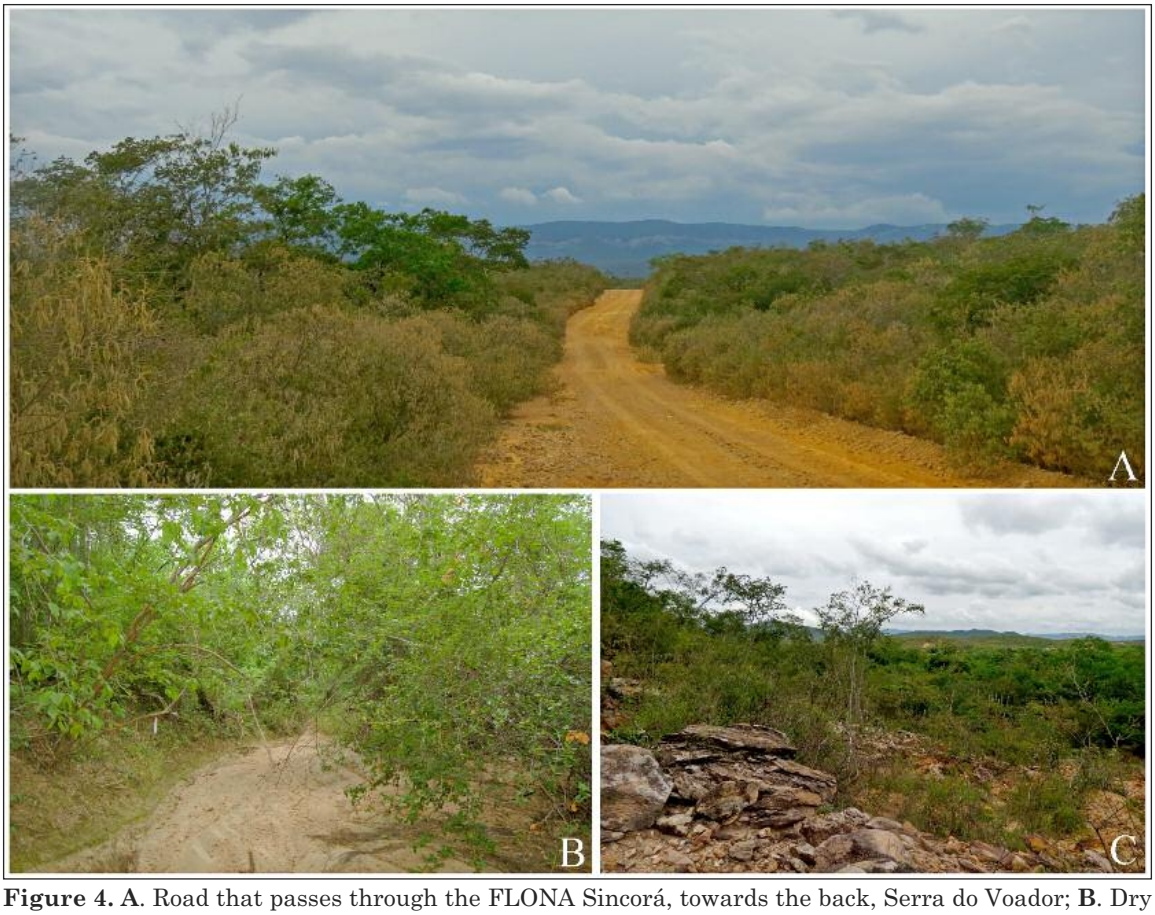


Figure 4. A. Road that passes through the FLONA Sincorá, towards the back, Serra do Voador; **B**. Dry bed of the Rio Combuca; C. Rock outcrop at the Serra do Voador.

Photographs: Grênivel Costa

11. *Tacinga* Britton & Rose

11.1. *Tacinga funalis* Britton & Rose, *Cact*. **1**: 39–40 (1919).
Common names: quipá-voador, trança-perna, rabo-de-rato.

Shrubby liana to 12m long., with decumbent terminal branches; stems sparsely branched, cylindric, 30–100×1–1.5cm, epidermis grey-green to purplish; areoles 1mm diam., spirally arranged along the stem, 0.3–1cm apart, sunken and punctiform, unarmed, with white to brown trichomes, glochids translucent. Flowers subapical, c. 8×2cm; pericarpel narrowly turbinate, 3×5 mm, grey-green, covered in areoles; outer perianth-segments 1×0.5–1cm, narrowly ovate, acute at apex, purplish; inner segments c. 7, 2–4cm long, acute and recurved at apex, greenish to cream-white; stamens erect, exserted in relation to the perianth-segments, 17mm long, anthers reddish; style 45mm long, stigma-lobes 4–6, spreading above the stamens. Fruit not observed.

Endemic to northeastern Brazil, it occurs in Bahia, Pernambuco and Piauí, growing in caatinga associated with gneissic and granite outcrops (Taylor & Zappi, 2004; Zappi *et al*., 2015). Found within three protected areas in Bahia, *Tacinga funalis* is known from the Gruta dos Brejões, Parque Estadual de Morro do Chapéu and Parque Nacional do Boqueirão da Onça, Chapada Diamantina (Taylor & Zappi, 2004). This peculiar species of *Tacinga* has a climbing habit shared only with the Vulnerable species *Tacinga braunii* Esteves Pereira, from northeastern Minas Gerais.

At the FLONA Sincorá it was found growing at 200–400m alt., in tall, well preserved caatinga. However, only few individuals (c. 7) were found, with a fragmented distribution along the track to the Rio Combuca and the track that cuts across the protected area (Figure 4). Most specimens seen were sterile, with a single flowering specimen found in April.

Examined material: Bahia, Mun. Contendas do Sincorá, Floresta Nacional Contendas do Sincorá (FLONA), 13°55'29.8"S, 041°04'30.1"W, 4 Apr. 2014, *M.B. Medeiros et al.* 604 (HURB); l.c., 24 Mar. 2013, *M.O.T. Menezes* 313 (EAC).

11.2. *Tacinga inamoena* (K. Schum.) N.P. Taylor & Stuppy in *Succ. Pl. Res*. **6**: 119 (2002).
Common name: quipá.
Fig. 3G.

Shrub or subshrub to 0.5(–1) m tall; stems well branched, all flattened and elliptic, rounded or obovate in outline, 4–12×3–7.5cm, epidermis bright to grey-green; areoles 2mm diam., arranged in a network over the stem, 0.8–1.5cm apart, unarmed, sunken and punctiform, glabrous, glochids translucent. Flowers subapical at the edge of the branches, 4–6×3.5–4cm, opening during the day; pericarpel globose, c. 3×3mm, green, with glochidiate areoles; perianth-segments red to orange, inner segments c. 2–3×2–2.5cm, ovate to oblong-ovate, reflexed at apex; stamens erect, exserted in relation to the perianth-segments, 20mm long, anthers reddish; style 30–40mm long; stigma-lobes 5–6, spreading above the stamens. Fruit globose, 1.7–2.5×2.3cm, green turning yellow to reddish when ripe; seeds 10–20 per fruit, lenticulate, pale brown, 4–5×2–3mm.

This Brazilian endemic is found in open caatinga and rock outcrops throughout northeastern Brazil, reaching the north of Minas Gerais and occurring on a wide variety of substrates and at altitudes from sea level to 1550m (Taylor & Zappi, 2004; Zappi *et al*., 2015; Menezes *et al*., 2011). *Tacinga inamoena* is amongst the commonest species in the region and is easily identifiable by its elliptic to obovate, apparently unarmed stems with red to orange, showy flowers. This species is able to tolerate some habitat disturbance and today its fruits are possibly dispersed by cattle.

At the FLONA Sincorá this plant is abundant and was commonly found between 200–350m alt., in and near gneissic rock outcrops and on the stony floor in shrubby caatinga. It was found with flowers and fruits between February and April.

Examined material: Bahia, Mun. Contendas do Sincorá, Floresta Nacional Contendas do Sincorá (FLONA), Fazenda do Pastor, caatinga, 13°53'58"S, 41°03'48"W, 23 July 2014, *M.R. Peixoto et al*. 3 (HURB); l.c., ramal contrário à entrada para a sede da FLONA, 13°55'46"S, 41°05'16"W, 6 Feb. 2015, *L.Y.S. Aona et al*. 3963 (HURB); l.c., lado direito da BR-026 em direção a fazenda Vargem Grande, 13°45'46.1"S, 41°02'28.0"W, 11 May 2012, *V.V.F. Lima et al.* 6 (CEN).

11.3. *Tacinga palmadora* (Britton & Rose) N.P. Taylor & Stuppy in *Succ. Pl. Res.* **6**: 112 (2002).
Common names: palmatória, quipá-de-espinho.
Fig. 3H.

Shrubs reaching 2(–4)m tall; stems well-branched, all flattened and obovate to elliptic in outline, 9.5–13×4.5–5cm, epidermis glaucous, grey-green; areoles 2mm diam., arranged in a network over the stem, 1.5–3cm apart, prominent, spines (0–)1–10 per areole, acicular, pungent, initially golden, pale grey when old, central spines 1–2, 1–3cm, radials 7–8, 0.4–1.5cm, glochids abundant, translucent. Flowers subapical at the edge of the branches, 1.5–2.7×2–3cm, opening during the day; pericarpel cylindric, c. 2×3 mm, green, with glochidiate areoles; perianth-segments red, outer segments oval, 1–1.3×1–1.5cm, inner segments 2.5–3×2.5cm, ovate to oblong-ovate, reflexed at apex; stamens erect, exserted in relation to the perianth-segments, c. 15mm long, anthers yellow; style 22mm long; stigma-lobes 4–5, spreading above the stamens. Fruit ovoid to subglobose, 2–3cm diam., 2.5–3×2cm, green turning red or purplish, glochidiate, funicular pulp translucent; seeds 3–4 per fruit, lenticulate, c. 5mm long.

Endemic to northeastern Brazil, this species occurs in the states of Ceará, Sergipe, Alagoas, Pernambuco, Paraíba, Rio Grande do Norte and Bahia (Zappi *et al*., 2015), occurring in caatinga, agreste and carrasco vegetation between 200 and 1020m alt. (Taylor & Zappi, 2004; Menezes *et al*., 2011). This species is found in several protected areas in Bahia, such as the Parque Estadual de Morro do Chapéu, Parque Nacional da Chapada Diamantina and others (Taylor & Zappi, 2004). It is not possible to exclude the occurrence of a less common species, *Tacinga werneri*. Vegetatively and in flower the two species are very similar but the fruit of *T. werneri* is white and egg-shaped when ripe and the funicular pulp surrounding the seeds is bright pink, while *T. palmadora* has somewhat shrivelled, green to red or purple fruit, totally green inside.

At the FLONA Sincorá *T. palmadora* was relatively common, found at 200–600m alt., in shrubby caatinga, flowering during the dry season (July).

Examined material: Bahia: Mun. Contendas do Sincorá, Floresta Nacional Contendas do Sincorá (FLONA), Fazenda do Pastor, 13°53'58"S, 41°03'48"W, 22 Jul. 2014, *M.R. Peixoto et al*. 2

(HURB); l.c., Trilha via Rio Combuca, 13°55'05"S, 41°06'53"W, 6 Feb. 2015, *L.Y.S. Aona et al.* 3960 (HURB).

Conclusions and recommendations

Despite intensive studies of the Cactaceae in eastern Brazil, the situation of the family within officially protected areas is not very well understood. Due to the importance of cactus species' ecological role in the Brazilian Caatinga vegetation, information regarding the extant populations of Cactaceae in protected areas of northeastern Brazil is a fundamental tool for directing conservation actions such as habitat monitoring, management and recovery, especially for endemic and endangered species. However, recording individuals within the areas is not enough to maintain population viability. Sometimes the species are already represented by a very low number of individuals, as was the case in the studied area with *Brasiliopuntia brasiliensis*, where a single specimen was found, or *Hylocereus setaceus*, where only two individuals were seen. We wonder how sustainable the future of the majority of the species studied will be if the major part of these species' distributions are confined to the protected areas and yet the populations are either extremely small and/or severely fragmented. It may be interesting to consider an informed reintroduction of these species from local stock in order to recover their populations within the protected area.

Arrojadoa penicillata, *Brasilicereus phaeacanthus*, *Pereskia aureiflora* and *Stephanocereus leucostele* are all currently affected by destruction of the Caatinga (Zappi *et al.* 2011). Apart from those not currently regarded as endangered taxa, *Pereskia aureiflora* and *Espostoopsis dybowskii* are considered to be Endangered (Zappi *et al.*, 2011, IUCN, 2015). These species deserve special attention within the protected area and efforts are being made to monitor and possibly include further populations of these species within the current management plans that are being developed for the FLONA. The immediate problems that affect this protected area are: the presence of cattle that causes further deterioration of the vegetation; and development plans to enlarge the track that runs within the reserve that particularly threatens the only population of *Melocactus concinnus* found within the area.

The Caatinga and its ecotone with the Atlantic Rainforest, known locally as agreste or mata-decipó, were included as priority areas or 'Hotspots' for cactus conservation in the National Cactaceae Action Plan (Ribeiro-Silva *et al.*, 2011) suggesting milestones and conservation actions for the conservation of this charismatic group of plants, such as increasing species studies and information dissemination, their re-introduction/reinforcement in protected areas and the subsequent monitoring of the populations.

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